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Review

Mercury in rice paddy fields and how does some agricultural activities affect the translocation and transformation of mercury - A critical review

Zhenya Tang [a,c], Fangling Fan [b,**], Shiping Deng [e], Dingyong Wang [c,d,*]

[a] Faculty of Environmental Science & Engineering, Kunming University of Science &Technology, Kunming, China
[b] School of Energy and Environment Science, Yunnan Normal University, Kunming, China
[c] Interdisciplinary Research Center for Agriculture Green Development in Yangtze River Basin, Southwest University, Chongqing, China
[d] State Cultivation Base of Eco-agriculture for Southwest Mountainous Land, College of Resources and Environment, Southwest University, Chongqing, China
[e] Department of Plant and Soil Sciences, Oklahoma State University, Stillwater, OK, USA

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ABSTRACT

Human exposure to methylmercury (MeHg) through rice consumption is raising health concerns. It has long been recognized that MeHg found in rice grain predominately originated from paddy soil. Anaerobic conditions in paddy fields promote Hg methylation, potentially leading to high MeHg concentrations in rice grain. Understanding the transformation and migration of Hg in the rice paddy system, as well as the effects of farming activities, are keys to assessing risks and developing potential mitigation strategies. Therefore, this review examines the current state of knowledge on: 1) sources of Hg in paddy fields; 2) how MeHg and inorganic Hg (IHg) are transformed (including abiotic and biotic processes); 3) how IHg and MeHg enter and translocate in rice plants; and 4) how regular farming activities (including the application of fertilizer, cultivation methods, choice of cultivar), affect Hg cycling in the paddy field system. Current issues and controversies on Hg transformation and migration in the paddy field system are also discussed.

1. Introduction

Mercury (Hg) is a global contaminant because it can travel long distance through the atmosphere, resulting in wide distribution of Hg in water, air, and soil (Fitzgerald et al., 2007; Nriagu, 1979; Schroeder and Munthe, 1998). Concerns on the toxicity of mercury was intensified following the deaths of hundreds of people caused by Minamata disease in Japan in the 1950s (Harada, 1995; Matsumoto et al., 1965). Since then, the hazards of mercury have been addressed by scientists around the world. Recently, the Minamata Convention on mercury was approved by the government of 120 countries and regions, with the holding of the first meeting of its Conference from 24 to 29 September 2017 in Geneva, Switzerland (www.mercuryconvention.org). Multiple factors contribute to the scope and range of Hg damage to biological systems. Since Hg can undergo long-range transport, harm can occur in areas distant from the source (Lawrence et al., 2013; Paige et al., 2013). Furthermore, Hg can be transformed into highly virulent methylmercury (MeHg) by microorganisms or other factors (Ullrich et al., 2001), and undergo bio-amplification along food chains (Peng et al., 2015). Generally, methylation of Hg is by anaerobic microorganisms, thus the primary concern was mainly focused on aquatic ecosystems, especially in fishes, seashells, and sea mammals (Peng et al., 2015). Several acute mercury poisoning incidents were indeed caused by consumption of aquatic products (Peng et al., 2015). Similarly, higher MeHg exposure risks can be associated with the consumption of rice (*Oryza sativa* L.) grain in areas with high Hg background levels (Meng et al., 2010, 2011, 2012). Initial studies of mercury in rice grain mainly focused on the effects of mercury-containing pesticides on total Hg in grain (Rahman and Gurcharn Singh, 1946; Smart and Hill, 1968; Smart and Hill, 1969). Interest lessened when mercury-containing pesticides were banned. However, Hg levels in rice grain regained attention when a high bioaccumulation factor of MeHg in rice grain was reported in the Guizhou Hg pollution area (Feng et al., 2008, 2010; Horvat et al., 2003b).

As one of the most important crops, rice and its byproducts provide staple foods to more than half of the world's population (Zhang et al., 2010a, 2010b), despite only about 10.7% of fields (about 160 million ha) being cultivated with rice (Faostat, 2014)). Of those paddy fields, approximately 89% utilize flooding (Sohrabi et al., 2012). Flooded

* Corresponding author. College of Resources and Environment, Southwest University, Chongqing, 400715, China.
** Corresponding author.
E-mail addresses: zytang@kust.edu.cn (Z. Tang), fangling.fan@outlook.com (F. Fan), shiping.deng@okstate.edu (S. Deng), dywang@swu.edu.cn (D. Wang).

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conditions can promote the growth of sulfate-reducing bacteria (SRB), iron-reducing bacteria (FeRB) (Kerin et al., 2006), methanogens, and other anaerobic microorganisms. Activity of these anaerobes contributes to higher MeHg concentrations in rice grain compared with other crops, and increased risk of MeHg exposure to human beings (Feng et al., 2008; Feng et al., 2010).

Transformation and migration of mercury in rice paddies are determined by many factors, including the forms of mercury (Liu et al., 2019a), soil microorganisms (Hu et al., 2013; Li et al., 2019; Liu et al., 2019b), rice varieties (Rothenberg et al., 2012), farming activities (Tang et al., 2018, 2019; Wang et al., 2014b), and other factors (Nakanishi et al., 2006; Zhang and Hsu-Kim, 2010). Integrating studies from multiple disciplines can address questions such as: What are the sources of mercury in the paddy field system? How is mercury transformed in paddy soil? How is methyl mercury taken up by the rice plant and translocated to grain? Therefore, this review examines the current state of knowledge on the transformation of Hg in the paddy field system, uptake and translocation of Hg in rice plants, and the effects of farming activities on these processes. While much has been learned about Hg in the rice paddy system, we also explore potential avenues for future research to address mitigation of deleterious levels of environmental contaminants.

2. Sources and transformation of mercury in rice paddy fields

Mercury is a heavy metal that can circulate in the rock-soil-water-atmosphere biosphere with different forms. Soil parent material is the primary source for rice paddy field soil. Soil is the largest Hg reservoir globally, containing about 1088 ± 379 Gg of Hg in the top 20 cm of soil (Li et al., 2009; Liu et al., 2019a; Smith et al., 2008; Tang et al., 2019; Wang et al., 2019a). In addition to soil parent material, atmospheric deposition (Sun et al., 2019a), chemical industry emissions (Yasuda et al., 2004), metal mining and smelting (Zheng et al., 2007), wastewater irrigation (Navarro et al., 1996; Wagner-Döbler, 2003), and application of fertilizer or pesticides (Mortvedt, 1995; Smart and Hill, 1968; Tang et al., 2018; Zhong et al., 2018) are considered as the main external pathways of Hg entering the rice system (Fig. 1).

2.1. Mining, smelting and chemical industries

Metal mining and smelting, as well as chemical industry activities have contributed to acute Hg contamination in many areas of the world. For example, Minamata disease was caused by the Hg in the discharge from chemical fertilizer plants. Additionally, many reports of exposure to MeHg from rice consumption originate in metal mining and smelting areas.

The concentrations of MeHg and THg in some metal mining, smelting and chemical industries are listed in Table 1. Average Hg concentrations in Hg or gold mining areas (Table 1) were far more than the soil background levels of China (11 μg kg^{-1}) or the U.S. (10 μg kg^{-1}) (Wang et al., 2019a). Even within a metal mining region, the uneven distribution of mining activities leads to wide differences in Hg concentration and form. For example, THg and MeHg concentrations could be as high as 4400 mg kg^{-1}, and 67 μg kg^{-1}, respectively, in paddy soil from the Wanshan Hg mine area in China (Table 1), and as low as 0.1 mg kg^{-1} THg and 0.1 μg kg^{-1} MeHg.

The effects of mining, smelting and chemical industries are mediated through the atmosphere, water runoff, and leaching. For example, studies showed that the Hg flux in paddy fields near Wanshan and Yanwuping Hg mining area, China could be in the range of 18–234 ng m^{-3} (Zhang et al., 2010b) or 12–180 ng m^{-3} (Qiu et al., 2013a). The concentrated distribution of chemical industries and minerals in Southern Asia has led to Hg concentrations greater than 80 mg kg^{-1}, mainly in surface soils of this region (Wang et al., 2019a). Given that rice is mainly cultivated in Asia, the high Hg concentrations in Southern Asian soils could pose a significant risk of Hg exposure from rice consumption (Qiu et al., 2005b).

2.2. Atmospheric deposition

Unlike other heavy metals that exist in the atmosphere for short periods in the particulate phase, Hg can remain in the atmosphere in the gaseous elemental form (>95% of total gaseous Hg) for 0.5–2 years, resulting in transport with the atmosphere all around the world (Feng, 2005). Atmospheric transport of Hg and its eventual deposition contribute to background levels in areas distant from sources. The deposition of Hg is correlated with the direction of atmospheric movement, but more importantly with the distance from the Hg emission sources, such as coal-fired power plants, chlor-alkali and battery production facilities, and metal mining and smelting operations (Feng, 2005). For example, the Hg concentration of ambient air downwind of a coal-fired power plant was much higher than background Hg levels of

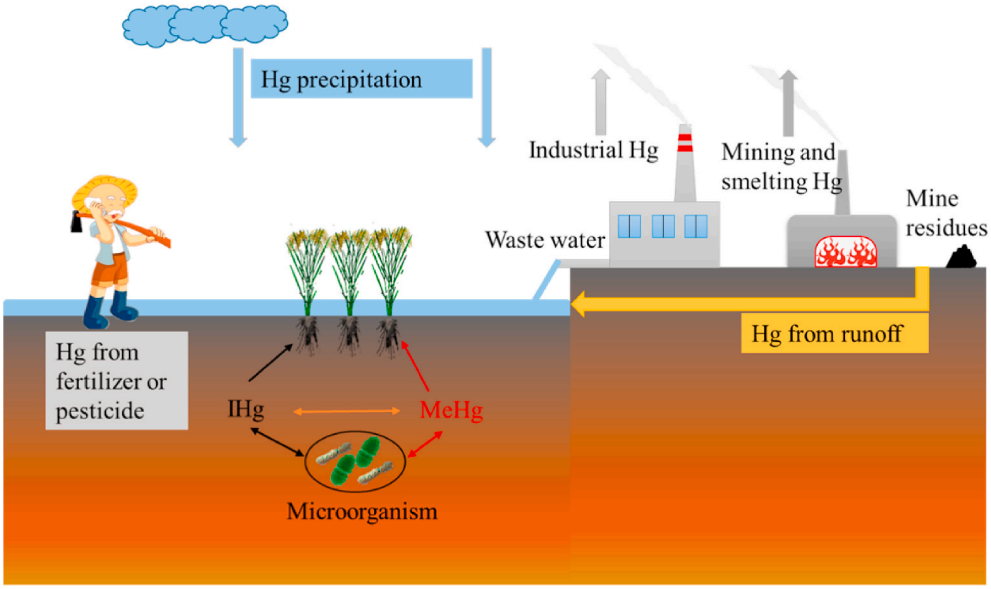


Fig. 1. The sources of Hg in rice paddy fields. Except for soil parent material, atmospheric deposition, emissions from chemical industries, metal mining and smelting, wastewater irrigation, and application of fertilizer or pesticide are the main external pathways of Hg into the rice paddy system.

Table 1
The THg and MeHg concentrations of soil and rice grain and THg deposition volume in mining and smelting areas.

Location	Soil THg	Soil MeHg	THg in Rice grain	MeHg in Rice grain	Reference
	mg kg^{-1}	μg kg^{-1}	μg kg^{-1}	μg kg^{-1}	
Wanshan Hg mining area, China (7)	0.51–130	1.7–21	130–210	15–30	Li et al. (2014)
	0.25–354	1.27–67	11.1–569	8.03–144	Horvat et al. (2003a)
	0.1–4400	0.1–20			Qiu et al. (2005a)
	45–130	7.6–21	130–210	15–30	Li et al. (2014)
	0.29–311	0.27–12.1	8.63–508.3	1.19–44.3	Zhang et al. (2010b)
	27.9–294.89	1.7–16.6			Meng et al. (2014b)
	15.41–34.93	1.6–10.4			Meng et al. (2014b)
Wukeng and dashuxi, China (1)	31–527	0.8–10	24.8–548	3.8–18	(Dai et al., 2012; Meng et al., 2010; Rothenberg et al., 2012)
Gouxi Hg mining, China (1)	13–36	2.6–13	128–622	18–62	(Meng et al., 2010; Rothenberg et al., 2012)
Xunyang Hg mining area, China(2)	5.4–120	1.2–11	50–200	8.2–80	Qiu et al. (2012a)
	0.31–121	0.24–8.9	4.5–71	2.1–59	Ao et al. (2017)
Wuchuan Hg mining area, China (2)	13–2100	0.093–1.1	6.0–113	3.8–12	(Li et al., 2008, 2012)
	0.33–320	0.69–20	9.1–570	4.2–18	Qiu et al. (2006a)
Yanwuping mining area, China (1)	0.24–240	0.34–7.3			Qiu et al. (2013b)
Danzhai, China (1)	7.66–33.89	1.9–3.8	85.8 + 43.6	20.8 + 9.3	Meng et al. (2014b)
Lanmuchang, China (1)	1.9–15.98	2.2–13.1	22.0 + 5.3	12.3 + 4.9	Meng et al. (2014b)
Fankou, China (1)	0.59–5.52	0.3–8.5	16.0 ± 4.1	2.0 + 0.5	Meng et al. (2014b)
Baojing, China (1)	28.99–35.91	3–6.2			Meng et al. (2014b)
Chashula, China (1)	50.78–85.69	4.5–10.9			Meng et al. (2014b)
Jijian, China (1)	138.05–172.47	6.5–9.8			Meng et al. (2014b)
Niudouping, China (1)	6.83–16.71	2.8–6.3			Meng et al. (2014b)
Tongren, China (1)	162.99–217.34	4.7–9.1			Meng et al. (2014b)
Huaxi, China (1)	0.24–0.38	0.24–0.58	3.8–10.3	1.8–4.5	Meng et al. (2010)
Zhanghe, China (1)			2.5–5.0	0.69–1.5	Rothenberg et al. (2011)
Qingzhen Hg polluted area (1)			2.53–33.5	0.71–28	Horvat et al. (2003a)
Hg mines, Alaska, USA (1)	0.03–5326	0.02–41			Gray et al. (2000)
Lombok gold mining, Indonesia (1)	103–8364			11–115	Krisnayanti et al. (2012)
Almaden Hg mine, Spain (1)	3.4–8889				(Higueras et al., 2003; Nevado et al., 2003, 2009)
Idrija Hg mine, Slovenia (1)	0.18–2759	0.32–80			(Hines et al., 2000; Horvat et al., 2002; Kocman et al., 2011)
palawan Hg mine, Philippines (1)	0.0012–566				(Cortes-Maramba et al., 2006; Gray et al., 2003)

air in the northern hemisphere, and surface soil around the coal-fired power plant contained 8.5 times of the background Hg level for soils in Guizhou province, China (Gao et al., 2018b).

It has been estimated that approximately 3200 Mg of Hg is deposited into the ground world-wide each year (Jiskra et al., 2015; Mason et al., 2012). A considerable portion of Hg deposition into crop fields takes place in Asia, especially China due to being the largest atmospheric Hg emission area in the world. For example, THg deposition through rainfall and dry deposition in Gouxi, China were 365–4760 μg m^{-2} yr^{-1} and 12–1652 μg m^{-2} yr^{-1}, respectively (Meng et al., 2010; Rothenberg et al., 2012). These rates were 1.02–87.5 and 58.5–584.0 fold more than Hg deposition in northeastern United States of American (Yu et al., 2014). The deposited Hg is mostly in the forms of Hg^0, reactive gaseous Hg, and particulate Hg, which can be transformed to MeHg by Hg-methylating microorganism in paddy soil (Zhao et al., 2016). Therefore, Hg deposition into paddy fields may increase MeHg concentrations, compared with other Hg sources.

2.3. Wastewater irrigation

To help meet the increasing demand for water as the world population grows, irrigation with treated wastewater has expanded both in developing and developed countries (WHO, 2006; Zolti et al., 2019). Wastewater irrigation leads to heavy metals entering the crop field along with high concentrations of nitrogen, phosphorus, potassium, and other nutrients. Studies showed soil Hg concentrations in wastewater irrigation areas were significantly higher than in the background area (Guédron et al., 2014). Similarly, Hg concentrations in rice grain from wastewater irrigation areas were greater than the national health standard of China (20 μg kg^{-1}) (Zhang et al., 2010a). The high concentration of Hg in wastewater, on one hand, could be directly absorbed by the rice plant, leading to the high Hg concentration in rice plants (Guédron et al., 2014). On the other hand, the wastewater could increase soil Hg concentrations and change the soil microorganism community structure, ultimately leading to increased Hg concentrations in rice plants (Chanda et al., 2011). Additionally, little oversight of Hg concentrations in wastewater may lead to Hg concentrations excessing the national health standard of China (Liu, 1994). Therefore, wastewater irrigation could bring mercury pollution harm to agricultural fields. It is also increasingly common for urban green belts to be irrigated with wastewater after simple purification. Clearly, full assessment of Hg exposure risks to humans from agriculture fields and urban areas irrigated with wastewater requires further study.

2.4. Application of fertilizer and pesticides

Applications of fertilizer and pesticides are regular methods for modern agricultural production. However, they can increase the risk for human exposure, as exemplified by the paddy field system. Some fertilizers and pesticides contain measurable amounts of Hg (Table 2), leading to significant increases in Hg concentrations in paddy soil after long term fertilizer or pesticide use (Tang et al., 2018). On the other hand, some fertilizers or pesticides can increase the availability of Hg and promote migration into rice plants (Liu et al., 2016b; Tang et al., 2018). This topic will be explored in greater detail in sections 4.1 and 4.2. Generally, phosphorus fertilizer (0.0 mg kg^{-1} - 41.3 mg kg^{-1}) and organic amendments or fermented urban sludge (0.0 mg kg^{-1} – 2.0 mg kg^{-1}) usually lead to greater soil Hg concentrations, compared to urea (2.10 μg kg^{-1}) or potassium chloride (not detected), because of its higher Hg concentration (Table 2). Additionally, the hydroxide radical, carboxyl, and hydrosulphonyl compounds from organic amendments could increase the solubility of Hg in paddy soils by complexing and/or coordinating with Hg (Barkay et al., 1997; Miskimmin et al., 1992; Wang et al., 2018).

2.5. Mercury forms transformation in paddy soil

2.5.1. The methylation of IHg in paddy fields

In nature, mercury methylation takes place in both biological and non-biological pathways, of which biomethylation is the primary

Table 2
The concentration of THg in fertilizer and pesticides.

Fertilizer or pesticide	THg concentration[c]	Reference
Urea	2.10[a]	Zhao and Wang, 2010
Ammonia sulfate	0.17[b]	Zhao and Wang, 2010
Ammonia chloride	ND[b]	Zhao and Wang, 2010
Potassium chloride	ND[b]	Zhao and Wang, 2010
Ammonium phosphate	0.0–0.7[b]	(Feng et al., 2009; Zhao and Wang, 2010)
Calcium superphosphate	0.1–41.3[b]	(Feng et al., 2009; Tang et al., 2018; Zhao and Wang, 2010)
NPK compound fertilizer	0.07–1.2[b]	(Carbonell et al., 2011; Zhao and Wang, 2010)
Organic amendments	0–2[b]	Feng et al. (2009)
Swine manure	4.88[a]	Tang et al. (2018)
Municipal solid waste	1.68[b]	Carbonell et al., 2011
Insecticide	0.8–3.19[b]	Campos, 2003
Fungicide	1.74–3.47[b]	Campos, 2003
Herbicide	1.21–1.75[b]	Campos, 2003

[a] represents μg kg^{-1}.
[b] represents mg kg^{-1}.
[c] ND indicates not detected.

process (Raposo et al., 2008). The non-biomethylation reaction is mainly a photochemical reaction in the atmosphere with Hg^0 and CH_3-X (X is a halogen atom) producing MeHg (Castro et al., 2011), and in aqueous systems with Hg^{2+}, Hg^{1+}, Hg^0 and iodide methyl (CH_3I) (Yin et al., 2014). The biomethylation of Hg is conducted by microorganisms containing a two-gene cluster *hgc*AB (Parks et al., 2013), including sulfate-reducing bacteria (SRB), iron-reducing bacteria (FeRB) (Kerin et al., 2006), methanogens, and other microorganisms such as fermentative, acetogenic, and cellulolytic microorganisms.

Paddy field is the largest artificial wetland system (Rothenberg et al., 2012, 2014). Long-term flooding or flooding-drying cycles make the paddy field system one of the most important Hg methylation areas. In the rice pot experiments, the SRB were considered as the main methylator in paddy soil because SRB inhibitors significantly decreased the production of MeHg (Tang et al., 2019). Some field surveys demonstrated that *Geobacteraea* and *Methanogen* are the main Hg-methylating microorganisms (Liu et al., 2014, 2016b). However, not all SRB, FeRB, and methanogens can methylate IHg into MeHg. Additionally, the efficiencies of Hg-methylating microorganisms are also quite different (Table 3). Highly efficient Hg-methylators have been found in FeRB (*Geobacter bemidjiensis Bem*), and SRB (*Desulfovibrio* sp. *X2, Desulfovibrio desulfuricans ND132, and Desulfomicrobium baculatum X*). The rates of Hg methylation by methanogens seem lower than FeRB and SRB. However, a recent study showed that MeHg production by methanogens rivals that of *Deltaproteobacteria* during batch culture growth (Gilmour et al., 2018b). Additionally, microbial communities not traditionally known for methylation, such as *Xanthomonadales* and *Chitinophagaceae*, may contribute significantly to MeHg concentrations in paddy soil (Liu et al., 2019b). The specific roles microorganisms play in mercury methylation in rice fields remains a topic for further study.

Mercury methylation in paddy soil is not only related to activities and quantities of Hg-methylating microorganism, but Hg concentration and forms. Studies showed that MeHg concentrations in paddy soil were positively correlated with THg concentration (Feng et al., 2010; Qiu et al., 2006b, 2008; Tang et al., 2018; Zhu et al., 2008). The Hg^0, $HgCl_2$ and β-HgS forms are much easier methylated into MeHg by microorganism compared with Hg-DOM, α-HgS, and nano-HgS (Liu et al., 2019a; Zhao et al., 2016).

2.5.2. The demethylation process in paddy field system

The MeHg concentration in paddy soil is determined by the balance between the processes of Hg methylation and MeHg demethylation.

Table 3
The rate of Hg methylation in some Hg methylating microorganisms.

Strain	Rate of Hg methylation	Reference
Geobacter bemidjiensis Bem	75.0%	Gilmour et al., 2013
Desulfovibrio sp. X2	62.0%	(Gilmour et al., 2011; Graham et al., 2012)
Desulfovibrio desulfuricans ND132	53.0%	(Gilmour et al., 2011; Graham et al., 2012)
Desulfomicrobium baculatum X	34.1%	Gilmour et al., 2013
Geobacter daltonii FRC-32	30.0%	Gilmour et al., 2013
Desulfonatronospira thiodismutans ASO3-1	29.6%	Gilmour et al., 2013
Desulfovibrio sp. T2	27.0%	(Gilmour et al., 2011; Graham et al., 2012)
Syntrophus aciditrophicus SB	20.50%	Gilmour et al., 2013
Desulfovibrio psychrotolerans	15.0%	Graham et al. (2012)
Desulfovibrio africanus "Benghazi"	13.0%	(Gilmour et al., 2011; Graham et al., 2012)
Desulfitobacterium metallireducens	11.9%	Gilmour et al., 2013
Desulfovibrio aespoeensis Aspo-2	11.0%	Graham et al. (2012)
Desulfovibrio alkalitolerans	9.0%	Graham et al. (2012)
Acetonema longum APO-1	7.8%	Gilmour et al., 2013
Dethiobacter alkaliphilus AHT 1	5.4%	Gilmour et al., 2013
Methanolobus tindarius	3.6%	Gilmour et al., 2013
Methanomethylovorans hollandica	3.0%	Gilmour et al., 2013
Desulfovibrio sulfodismutans ThAcO1	1.9%	Graham et al. (2012)
Desulfosporosinus acidiphilus SJ4	1.0%	Gilmour et al., 2013
Ethanoligenens harbinense YUAN-3	0.9%	Gilmour et al., 2013
Desulfitobacterium dehalogenans	0.6%	Gilmour et al., 2013
Desulfosporosinus youngiae	0.6%	Gilmour et al., 2013
Desulfovibrio intestinalis	<0.005%	Limper et al., 2008

Generally, photodegradation (Seller et al., 1996), microbial reduction (Hu et al., 2013), and microbial oxidation (Oremland et al., 1991) are considered the three main pathways of MeHg demethylation.

Photodegradation of MeHg is a process mediated by ultraviolet radiation (Lehnherr and St. Louis, 2009) and enhanced by the presence of hydroxyl radicals (Black et al., 2012), singlet oxygen (Hammerschmidt and Fitzgerald, 2010), and organic ligands (Zhang and Hsu-Kim, 2010). The shallow water, abundant anions (such as NO_3^-, Cl^-) and rich organic matter in the paddy field seem beneficial to the demethylation of MeHg. However, photo-demethylation of MeHg reduced surface water MeHg loads by only 6% over the growing season (Alpers et al., 2014), because the high-density rice leaves reduced the amount of UV light reaching the water surface.

Microbial pathways for the degradation of MeHg are distinguished by reductive demethylation and oxidative demethylation. The choice of these processes is to a large extent controlled by environmental factors. The reductive demethylation of MeHg plays more important roles in aerobic and high-concentration Hg pollution environments, while oxidative demethylation of MeHg is observed in anaerobic and low Hg concentration environments. Reductive demethylation of MeHg is mediated by the Hg resistance (*mer*) system in bacteria. An organomercury lyase enzyme (*mer* B) cleaves the C–Hg bond to generate CH_4, and mercury reductase (*mer* A) can reduce Hg^{2+} to Hg^0 (Schaefer et al., 2004). The mechanism of oxidative demethylation of MeHg by microorganisms is still unclear. Studies speculated it might be related to metabolism of methyl nutrient bacteria, including methanogens and SRB (Marvin-DiPasquale and Oremland, 1998; Oremland et al., 1991). The probable demethylation reactions in methanogens and SRB are represented by the following equations, respectively:

$$4CH_3Hg^+ + 2H_2O + 4H^+ \rightarrow 3CH_4 + CO_2 + 4Hg^{2+} + 4H_2$$

$$SO_4^{2-} + CH_3Hg^+ + 3H^+ \rightarrow H_2S + CO_2 + Hg^{2+} + 2H_2O$$

Similar to Hg methylation by microorganisms, the demethylation of MeHg may also be a detoxification mechanism used by microbes to convert Hg to forms easily excreted from the cells. For example, *G. Bemidjiensis Bem* not only has the ability to methylate IHg to MeHg, but also to demethylate MeHg to IHg, thus controlling the concentrations of IHg and MeHg in the reaction system (Lu et al., 2017).

The demethylation of MeHg was not only observed in paddy soil, but also in rice plants. Generally, transgenes of *mer* A in rice plants could induce demethylation of MeHg. It is noteworthy that recent studies found that rice plants without transgenes also could demethylate MeHg (Strickman and Mitchell, 2017; Xu et al., 2016). Strickman (Strickman and Mitchell, 2017) speculated that demethylation in rice could be affected by UV light, since illuminated straw lost approximately 45% of MeHg, compared to a 22% reduction in roots without light. However, the specific mechanisms remain unknown.

3. The uptake and translocation of Hg in rice plants

Mercury is highly toxic to plants, inhibiting seed germination, reducing the length and weight of roots and stems, and decreasing growth rates. The toxicity of Hg is mainly due to its strong affinity for –SH functional groups of essential biomolecules. For example, Hg interaction with –SH groups in enzymes could alter protein conformation and disrupt key metabolic pathways, such as energy transduction. Redox reactions involving mercury can increase the generation of reactive oxygen species (ROS) resulting in oxidative stress to the plant. To lower the toxicity of heavy metals, plants have evolved a series of mechanisms, including complexing it with phytochelatins and/or metallothionein.

Mercury is a non-essential element for rice plants. Thus, rice plants do not have specific channels or transporters for Hg uptake. The uptake and transport of Hg in the plant are possible using transporters or channels of other heavy metals or nutrients (Esteban et al., 2008; Patra and Sharma, 2000). Although the specific channels or transporters involving Hg uptake and translocation still remain unclear, several lines of evidence have confirmed that translocation of MeHg and inorganic Hg (II) are by different pathways, and are allocated to different rice plant tissues (Meng et al., 2012, 2014a; Rothenberg and Feng, 2012; Yin et al., 2013). For example, Zhang reported the bioaccumulation factors (BAFs = the Hg concentration in rice tissue/the Hg concentration in soil) for MeHg in rice grain were far greater than BAFs for IHg, regardless of whether or not the site was contaminated with Hg (Zhang et al., 2010b). Therefore, understanding the uptake and translocation of IHg and MeHg in rice plants is one key to developing strategies to decrease the concentration of Hg in rice grain.

3.1. The uptake and translocation of IHg in rice plants

Mercury is a special heavy metal that can remain in a gaseous state for extended periods. This property leads to the uptake of Hg by rice plants from the surrounding atmosphere as well as paddy soil (Ackerman et al., 2010; Jiang et al., 2012; Meng et al., 2014a; Qiu et al., 2008; Rothenberg and Feng, 2012; Rothenberg et al., 2011; Zhang et al., 2010b; Zhu et al., 2008). As mentioned above, there are no specific channels or transporters for Hg uptake in rice roots. One possible mechanism of IHg ingress into rice roots involves Hg (II) passing through Ca^{2+}, Zn^{2+}, Cu^{2+}, and other metal channels in the plant cell membrane. Mercury is then complexed with phytochelatins in the cytosol of root cells before transport into the vacuole (Fig. 2). In support of this model, Gupta and Chatterjee found that exposure of rice roots to a Hg (II) solution induced the opening of Ca^{2+} channels in root membranes (Gupta and Chatterjee, 2014). Krupp and coworkers reported that Hg in the rice root cytosol could chelate with phytochelatins to form the PC_2Hg, (des-Gly)PC_2Hg, and (Glu)PC_2Hg structures (Krupp et al., 2009). Subsequent studies revealed the involvement of ABC and other transporters in the movement of Hg(II) into the plant vacuole (Park et al., 2012). This mechanism also lead to the IHg uptake from rice root hardly transported into rice aerial parts.

Isotope experiments demonstrated that IHg in ambient air was another source for rice plants (Strickman and Mitchell, 2017; Yin et al., 2013). In contrast with Hg (II) uptake by rice roots, the IHg absorbed by rice leaves is mainly in the form of gaseous elemental Hg (Hg^0_g, which accounts for more than 95% of total gaseous Hg) entering through stoma (Feng et al., 2010; Laacouri et al., 2013; Manceau et al., 2018; Meng et al., 2012). Small amounts of Hg enter leaves through nonstomatal pathways in the forms of reactive gaseous Hg and particle-bound Hg (Laacouri et al., 2013). Studies demonstrated that Hg^0_g entering rice leaves was oxidized to Hg^{2+} through an unknown mechanism (Laacouri

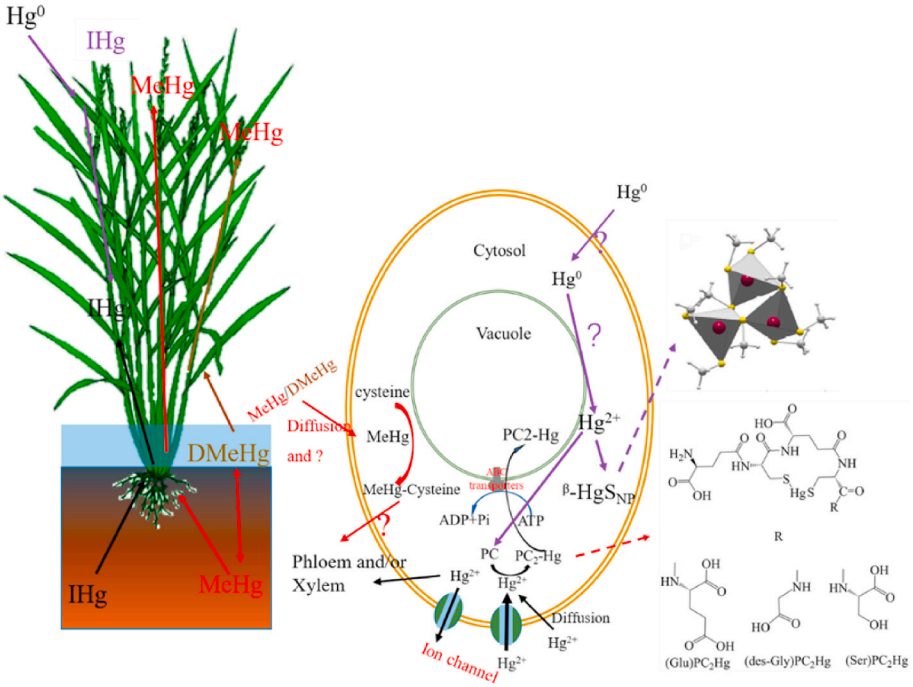


Fig. 2. The possible pathway of the uptake and translocation of IHg and MeHg in rice plants. Both IHg and MeHg enter rice plants through roots and leaves. The IHg into rice root is that the Hg^{2+} pass through diffusion or metal channels (Ca^{2+}, Zn^{2+}, Cu^{2+} *etc.*) of cell membrane into rice cell, and then are complexed with phytochelatins in cytosol of root cell, ultimately are transported into vacuole. Hg^0 entering rice leaves was oxidized to Hg^{2+} through some unknown approach, and then, in part, was complexed with thiolate to form PC_2–Hg, and more Hg was complexed with four sulfide or thiolate sulfur atoms to form metacinnabar-type-like (β-HgS) nanoparticulate. The MeHg and gaseous dimethyl mercury are absorbed by rice roots from paddy soil and by rice leaves from ambient air, and then are complexed with cysteine, ultimately are translocated into other tissues of rice plants.

et al., 2013), before being complexed with thiolate to form PC_2–Hg (Manceau et al., 2018). Moreover, Hg (>57%) had been shown to complex with four sulfide or thiolate sulfur atoms to form metacinnabar-type-like (β-HgS) nanoparticulate (Manceau et al., 2018).

The different mechanisms by which IHg is taken up from soil and the atmosphere lead to a differential distribution of IHg in rice plants. The general trend for IHg concentrations in rice tissues is: root > leaf > stalk > hull > grain (Meng et al., 2012). The IHg in rice roots mainly originates from paddy soil, while leaves, hull, and grain obtain IHg mainly from ambient air. Rice stalks contain IHg from both roots and leaves.

3.2. The uptake and translocation of MeHg in rice plants

Rice plants do not have the capacity to methylate IHg into MeHg (Meng et al., 2014a). Thus, the MeHg in rice plants originates from the surrounding environment. Paddy soils are considered the main source of MeHg for rice plants (Meng et al., 2014a, 2014b; Qiu et al., 2012b). Generally, MeHg is absorbed by rice roots from paddy soil, and then translocated to other plant tissues. A recent isotope experiment (Wang et al., 2018) demonstrated that 15.5%, 10.8%, and 8.5% of MeHg in rice grain, leaves, and upper stalk, respectively, were derived from the atmosphere as dissolved gaseous dimethyl mercury, while MeHg in roots originated primarily from paddy soil or water (Fig. 2).

It is not clear how MeHg enters rice plants and is translocated between tissues. In contrast with Hg (II) complexing with phytochelatins in rice roots, MeHg in rice root cells does not appear to complex with phytochelatins. Krupp and coworkers (Krupp et al., 2009) found no CH_3Hg-PC complexes in shoot or root extracts from rice plants cultivated in nutrient solution with CH_3Hg^+. The authors concluded that the two thiol bonds can more effectively chelate the divalent Hg^{2+} than the monovalent CH_3Hg^+ (Krupp et al., 2009). Additional studies with rice plants demonstrated that MeHg can complex with the –SH group in cysteine to form CH_3Hg-cysteine. Once complexed, MeHg-associated cysteine can be translocated like a mobile nutrient in rice plants (Meng et al., 2014a), leading to relative higher MeHg concentrations in grain than root, leaf, or stalk.

One possible explanation for different phenomenon between IHg and MeHg in rice plant (Fig. 3) is that the presence of Hg^{2+} could activate phytochelatins synthase, like Cd^{2+}. Specifically, two cysteine residues bind with one Hg^{2+} in phytochelatins synthase, bring them into closer proximity, and transfer them to the activation site in the N- terminal (catalytic domain) of phytochelatins synthase. The activated N-terminal domain catalyzes the transfer of the γ-Glu-Cys moiety of a molecule of GSH onto a second molecule of GSH, and eventually synthase the PCs (Cobbett, 1999, 2000). While CH_3Hg^+ just bind one cysteine residues, which cannot activate the phytochelatins synthase to catalyze the production of PCs. Additionally, the concrete process, involved enzyme and genes in rice plants are still needed further studies.

4. Effects of agricultural practices on the migration and translocation of mercury in paddy fields

Anthropogenic activities distinguish rice paddy systems from other wetlands. Agricultural practices employed in rice paddies, including cultivation and the application of fertilizers and amendments, can alter physicochemical properties, microorganism community structure, and Hg availability relative to native wetlands.

4.1. Organic fertilizers

Organic fertilizers such as animal manure and crop residuals improve soil and crop quality. The application of organic fertilizer can also affect Hg migration and transport in the paddy system. Three key features characterize the relationship between organic fertilizer application and Hg levels, forms, and availability.

First, organic fertilizers may lead to increase in the Hg content in

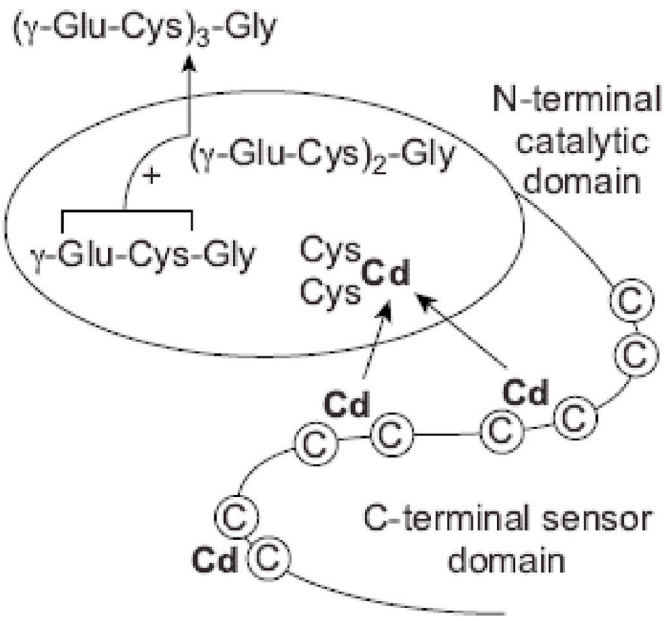


Fig. 3. A model for Cd $^{2+}$ in the function of PC synthase. The C-terminal domain of PC synthase acts as a local sensor of heavy metal ions, such as Cd. The Cys residues bind Cd ions, bringing them into closer proximity and transferring them to the activation site in the N-terminal (catalytic domain). The activated N-terminal domain catalyzes the transfer of the γ-Glu-Cys moiety of a molecule of GSH onto a second molecule of GSH or an existing PC_n molecule to form a PC_{n+1} product. Reprinted with permission from (A family of phytochelatin synthase genes from plant, fungal and animal species). Copyright (1999) Elsevier.

paddy soils, as discussed in section 2.4. Moreover, the Hg in organic amendments are predominantly in forms that are easily transformed into MeHg.

Second, the application of organic fertilizers could stimulate the growth of Hg-methylating microorganisms (Liu et al., 2019b; Windham-Myers et al., 2009, 2014). Under anaerobic conditions, methanogens can decompose organic matter and produce acetic acid, which promotes the growth of SRB (Reis et al., 1992) and FeRB (Marozava et al., 2018).

Third, IHg or MeHg can be bound by organic matter (OM), including dissolved organic matter (DOM), colloidal organic matter (COM), and particulate organic matter (POM). Organic matter has a variety of coordination functional groups, such as S-, O-, and N-containing moieties, in particular, reduced S groups (e.g., thiols, sulfides, and disulfides) (Skyllberg, 2008). These interactions are likely to dominate the chemistry of Hg species in many freshwater systems (Ullrich et al., 2001). The mobility and bioavailability of Hg in paddy soil and water decreases when bound to POM, and increases when binding COM or DOM. Given the rich DOM in natural water and paddy water, more than 90% of inorganic Hg (II) and 70%–97% of MeHg in water may be associated with DOM in fresh water, instead of soil or sediment particles (Hesterberg et al., 2001; Ullrich et al., 2001; Xia et al., 1999). DOM increased Hg bioavailability and promoted the production of MeHg by slowing the particle growth of β-HgS to boost the access of Hg in Hg methylated microorganism (Graham et al., 2012), oxidizing Hg^0 to Hg(II) by thiols of DOM (Zheng et al., 2011), or producing ·OH photolysis of organic matter in sunlight (Zhang and Lindberg, 2001).

However, DOM does not always promote the production of MeHg, some studies report that DOM decreased the bioavailability of Hg by reduction of Hg (II) to Hg^0, and volatilization of Hg (Costa and Liss, 2000; Zheng et al., 2011). Organic fertilizer from different sources might lead to various roles in the availability and conversion of Hg. For example, use of swine manure leads to much lower MeHg concentrations

in rice paddy soil and rice grain than returning rice straw (Tang et al., 2018, 2019; Zhu et al., 2015). Rice straw consists mainly of cellulose, hemicellulose, lignin and crude protein, and releases a large amount of DOM of small molecular size, such as acetate upon decomposition (Marvin-DiPasquale et al., 2014). These compounds favor the growth of SRB and FeRB, thereby leading to higher methylating potential and MeHg production (Liu et al., 2016b; Zhu et al., 2015).

4.2. Chemical fertilizer

Application of chemical fertilizers is a common method to increase or maintain crop yield. The application of chemical fertilizers, however, leads to changes in soil physiochemical properties, microorganism community structure, and rice growth conditions, which can affect the migration and transformation of Hg in the paddy field.

4.2.1. Nitrogen, phosphorus, and potassium fertilizers

Nitrogen, phosphorus, and potassium fertilizers are the three main macronutrients for crops. According to the statistics of the U.N. Food and Agriculture Organization, the applied dosages of nitrogen, phosphorus, and potassium fertilizer were 253.0 million tons in 2014 (Faostat, 2014). The long-term application of nitrogen fertilizer can result in soil acidification (Barak et al., 1997; Guo et al., 2010; Liang et al., 2013; Tang, 1998), increasing the availability of Hg.

The application of nitrogen can change the rate of organic carbon mineralization (Zhang et al., 2017), which affects the growth of soil microorganisms, including Hg-methylating microorganism such as methanogens (Belay et al., 1984), FeRB (Yang et al., 2012), and microorganisms that do not methylate Hg (such as nitrobacteria, ammoniated bacteria). For example, Zhang and Strickman demonstrated that application of nitrogen fertilizer promotes the growth of nitrate-reducing bacteria which can compete with microbial methylators of Hg for labile carbon (Strickman and Mitchell, 2018; Zhang et al., 2018). The concentration of Hg in nitrogen fertilizer is very low (Table 2), suggesting that inputs of Hg through nitrogen fertilizers are also at a low level. Long-term fertilizer experiments verified that no significant differences in soil THg concentrations were observed between the application and non-application of nitrogen fertilizer (Rutkowska et al., 2015; Tang et al., 2018). However, the long-term nitrogen fertilizer study also found that MeHg concentrations in nitrogen fertilized rice grain were significantly higher than those in unfertilized grain (Tang et al., 2018), suggesting nitrogen may promote uptake of MeHg from paddy soil and translocation in rice plants. Potential interactions between nitrogen fertilization and Hg uptake and translocation in rice plants warrant further studies.

Application of phosphorus fertilizer can increase THg levels in paddy soil (Feng et al., 2009), yet reduce the absorption of Hg by rice roots (Yin et al., 2016). Phosphorus application can lead to significantly higher MeHg concentrations in rice grain compared to treatments that did not include phosphorus (Tang et al., 2018; Yin et al., 2016). Phosphorus can promote advanced rice tillering and heading (Bates and Lynch, 1996), which may lead to increased amounts of MeHg translocated to rice leaves and grain. Additionally, some studies demonstrated the absence of phosphorus could induce the forming of the iron membrane to decrease the uptake of IHg and MeHg by rice root (Chen et al., 2014; Gao et al., 2018a; Liu et al., 2004b). Therefore, it is necessary to further investigate the roles of phosphorus fertilizer in the cycling of Hg in the paddy field system.

The effects of potassium on Hg transformation are rarely reported, but the ligands of potassium fertilizers, such as SO_4^{2-} or Cl^- from K_2SO_4 or KCl, significantly influence methylation of IHg. Cl^- can participate in photochemical reactions involving mercury. For example, Cl^- can obstruct the demethylation of MeHg by forming CH_3HgCl instead of more easily degradable forms (CH_3HgOH, and CH_3Hg^+) (Lee et al., 2019; Sun et al., 2013). Sulfate, on one hand, can facilitate the production of MeHg in paddy soil by stimulating the growth of SRB (Gilmour et al., 1998; King et al., 1999; Li et al., 2019), and/or enhancing the bio-availability of HgS by forming polysulfide compounds when excess sulfide is present (Drott et al., 2007; Li et al., 2019). On the other hand, excess sulfate may limit IHg availability by increasing the fraction of extracellular charged Hg–sulfide complexes under reduced conditions (Mitchell et al., 2008). Additionally, some SRB can demethylate MeHg during the process of sulfate reduction as indicated in section 2.5.2. Observations from natural lakes showed that when sulfate concentrations were in the range of 10–300 μmol L^{-1}, or when sulfide concentration were as low as 10 μmol L^{-1}, methylation of IHg was optimal. Otherwise, Hg-methylation rates and/or MeHg concentrations were negatively related to dissolved sulfide concentrations or sulfate reduction rates (Benoit et al., 1998; Gilmour et al., 1998). However, paddy pot experiments showed that concentration of sulfate exceeding 1000 mg kg^{-1} can still promote the production of MeHg (Li et al., 2019). The specific effects of sulfate on Hg methylation in paddy soil needs to be further explored.

4.2.2. Selenium fertilizer

Selenium (Se), an essential trace element for human beings and animals, can mitigate the toxicity of many heavy metals by forming selenoenzymes, including glutathione peroxidase and thioredoxin reductase. In biological systems, selenols can readily replace thiols in amino acids because of the chemical resemblance to thiols, leading to IHg and MeHg complexing with selenol-containing biomolecules. Compared to thiols, binding between Hg and selenols is stronger (Carty et al., 1983; Sugiura et al., 1978). Therefore, Hg and MeHg complexes with selenols are more stable than their thiol analogs, that is, Hg–Se antagonism, resulting in effectively reducing IHg and MeHg in rice with the Se addition to soil (Wang et al., 2014a; Zhang et al., 2012; Zhu et al., 2014).

Different from de-toxicity in animals, the Se reducing the concentration of IHg and MeHg in rice plants might be through the following pathway.

First, the formation of HgSe prevents the migration and methylation of IHg in soils. This was evidenced by the observation of HgSe in soil and plant with X-ray absorption near-edge structure (XANES) or transmission electron microscopy linked with energy-dispersive X-ray spectroscopy (TEM–EDX) (Wang et al., 2016), and the maximum rate (V_{max}) of Hg uptake by rice root decreasing by 11%–39% after the addition of Se in paddy soil (Wang et al., 2014a).

Second, Se at the appropriate concentration can enhance the development of iron plaques and apoplastic barriers in the endodermis, hindering both IHg and MeHg uptake (Wang et al., 2014a; Zhou et al., 2007; Zhu et al., 2014). Selenium can promote development of iron plaques and decrease accumulation of Hg in rice grain at low concentrations (below 0.5 μg mL^{-1}), but increase Hg in rice grain at high concentrations (over 0.5 μg mL^{-1}) (Li et al., 2015).

Furthermore, the addition of Se could restrict Hg uptake into root cells through the symplastic pathway by lowering the activity of membrane transporters (Wang et al., 2014a). Selenium can also compete with Hg for specific binding sites, such as the thiol groups of cysteine in membrane proteins, to defend higher plants against abiotic stresses (Feng et al., 2013).

4.2.3. Silicon fertilizer

Silicon (Si), as a beneficial element for rice plants, can enhance photosynthetic ability, improve grain yield and quality, as well as decrease the toxicity of heavy metals in rice plants (Hernandez-Apaolaza, 2014; Inanaga et al., 2002; Meharg and Meharg, 2015). Additionally, Si can precipitate metal in rice plant (Neumann and Zur Nieden, 2001), which contribute to metal detoxification and serve as a metal source when required. For example, Si mitigated Cd phytotoxicity by significant decreasing Cd uptake by rice roots and reducing translocation from root to shoot. In addition, Si stimulated deposition of Cd into the endodermal cell walls of rice roots (Farooq et al., 2016; Shi et al., 2005). Silicon in the soil solution exists primarily in the form of

monosilicic acid (H_4SiO_4), which forms Hg–Si co-precipitates, and transforms activated Hg into a binding or residual state with low bioavailability. Thus, the MeHg concentration in paddy soil decreases by 56.4% compared with the treatments with silicon fertilizer (Yin et al., 2016).

In addition to direct effects on Hg availability, silicon fertilizers may also have an indirect influence on Hg uptake by rice plants. For example, Si fertilizer could increase the mass and activity of rice roots, improving translocation of nutrients into rice grain, which may, in turn, affect the uptake and translocation of IHg and MeHg.

4.3. Biochar

Biochar contains biological residues that have undergone combusted under low-oxygen conditions, yielding a product with applications in soil remediation and wastewater management. Likewise, biochar applied to paddy soil could adsorb IHg and MeHg, lowering their availability to rice plants (Shu et al., 2016a, 2016b; Wang et al., 2019b; Zhang et al., 2018). Except for physical adsorption, Hg tends to bind to S functional groups in biochar with high S levels, and to O and/or Cl functional groups in biochar with low S levels (Liu et al., 2016a). However, the application of biochar could also lead to an increase in MeHg concentration in paddy soil because biochar, on one hand, can inhibit MeHg uptake by rice plants (Gilmour, 2018; Shu et al., 2016b), and on the other hand, promotes the growth of SRB (Shu et al., 2016b). Additionally, biochar also enhanced the reduction and oxidation of some *Geobacter and methanosarcina* as electron shuttle (Saquing et al., 2016; Tong et al., 2014), which might increase the methylation of IHg in paddy soil by those Hg-methylated microorganisms. In addition, biochars derived from different raw materials (Srinivasan and Sarmah, 2015) or prepared under different conditions would result in different properties and subsequent effects on Hg methylation.

4.4. Cultivation methods

The cultivation methods used in the paddy field could lead to distinct differences in the physicochemical properties of paddy soils (Zhou et al., 2014), which would affect Hg methylation in paddy soil and translocation in rice plants (Rothenberg et al., 2011; Sun et al., 2019b).

Generally, continuous flooding in paddy field ecosystems leads to higher MeHg concentrations in paddy soil and rice grain compared with alternating wetting and drying or drying ecosystems (Peng et al., 2012b; Rothenberg et al., 2016; Sun et al., 2019b). For example, 49% of THg in rice grain was in the form of MeHg in flooded soil, and only 4.7% of THg was in the form of MeHg in aerobic soil (Peng et al., 2012b). The continuously flooded condition, on one hand, enhances the mobilization of THg due to the reduction of iron (Fe) and manganese (Mn) oxides, which would release adsorbed or co-precipitated Hg into the solution (Ullrich et al., 2001). On the other hand, the anaerobic condition promotes Hg methylation by stimulating the growth of anaerobic microbes, including SRB (Compeau and Bartha, 1985; Wang et al., 2014b) and FeRB (Kerin et al., 2006).

The uptake of MeHg occurs throughout the entire growth period of rice plants, with a peak during the tillering and heading stages (Rothenberg et al., 2011). Water management before rice heading can significantly decrease the MeHg in rice grain compared to water management after rice heading (Rothenberg et al., 2014). Additionally, water management affects the growth of rice plants, leading to variation in the distribution of MeHg in rice tissues (Wang et al., 2014b). Interestingly, short-term draining of rice paddy fields seems to pulse the production of MeHg during flooding (Marvin-DiPasquale et al., 2014). Most of the anaerobes are not dead, but dormant during the short-term draining period (Rothenberg and Feng, 2012). When paddy fields are re-flooded, anoxic conditions contribute to decomposition of organic matter and stimulate recovery of anaerobes within hours, leading the increases in sulfate and MeHg levels in soil (Kelly et al., 1997; Rothenberg and Feng, 2012).

Rice cultivation with different crop rotation also caused various uptake and translocation of heavy metal (Sun et al., 2019b; Wu et al., 2011). For example, rice rotation with oilseed rape could significantly decrease the concentrations of Cd and Pb in rice grain by 46–80% and 17–86%, respectively. Sun surveyed several crop rotation systems in Chongqing, China and found that MeHg concentrations in the paddy soil of a rape-rice rotation were significantly higher than in the soil of a vegetable-rice rotation, and a winter-rice system that involves draining water (Sun et al., 2019b). Unfortunately, the effects of different rotation systems on Hg accumulation in rice plants are rarely reported.

4.5. Rice cultivars

To date, more than 400,000 rice accessions have been collected in the public germplasm repositories of the world (Wei and Huang, 2019). The uptake and transformation of heavy metals in rice plants was greatly impacted by cultivar (Peng et al., 2012a; Rothenberg et al., 2012). For example, Hg-binding efficiency of root surfaces (Liu et al., 2004a; Mei et al., 2009) and translocation of Hg in plants (Krupp et al., 2009) are regulated by genes differentially expressed in rice cultivars.

Although choice of rice genotype was associated with MeHg concentration in white rice, this was not the case with IHg (Rothenberg et al., 2012). Considerable variations in THg and MeHg concentrations were observed in different grain genotypes (Peng et al., 2012a). The rice genes involved in uptake, tolerance, and transport of Cadmium (Cd) and Arsenic (As) have been well documented (Atwell et al., 2014; Chen et al., 2019; Hu et al., 2016; Huang et al., 2018; Liu et al., 2004a; Luo et al., 2018; Mei et al., 2009; Zhang and Wing, 2013). For example, the *OsIRT1, OsIRT2, OsNramp1, OsNramp5, OsZIP1, OsZIP6,* and *OsCd1* genes are involved in the uptake of Cd in rice roots (Ishimaru et al., 2012; Kavitha et al., 2015; Nakanishi et al., 2006; Takahashi et al., 2011; Yan et al., 2018). The *OsHMA2, OsHMA3, OsCCX2, OsZIP7,* and *OsLCT1* gene products function in transport of Cd from rice roots to shoots or grain (Hao et al., 2018; Sasaki et al., 2014; Takahashi et al., 2012; Uraguchi et al., 2011); while the *rgMT, OsPCR1, OsPCS2,* and *SISAP1*-genes confer Cd tolerance (Das et al., 2017; Wang et al., 2015a). Unfortunately, information about Hg regulation in rice plants is still limited. Studies on Hg regulation genes in rice plants focused on the role of the *Mer*A/B transgene in decreasing MeHg concentrations (Eapen and D'souza, 2005; Ruiz and Daniell, 2009). Additional studies reported that *OsTCTP* (Wang et al., 2015b), *RM1003*, *RM110*, and *RM405* (WANG et al., 2013) were involved in Hg tolerance. Understanding the genetic regulatory mechanisms of mercury uptake and transfer in rice will facilitate targeted cultivation and screening of rice varieties with low mercury concentrations.

5. Conclusion and future perspectives

The potential risk of MeHg exposure through consumption of rice grain, especially for residents in Hg-contaminated areas and pregnant women, has been well recognized. To address this problem, scientists have made multiple breakthroughs, such as the discovery of essential genes for methylation of IHg (Parks et al., 2013), and the distribution of Hg in rice grain (Meng et al., 2014a). However, concerns of transformation and migration of Hg in paddy systems remain, largely due to the complexity of the Hg cycle in the environment. Future research could be directly:

1) To reveal the role of non-Hg methylating microorganisms in the transformation of Hg in paddy soil. Methylation of IHg is mainly conducted by microorganisms with *hgc*AB genes (Gilmour et al., 2011; Parks et al., 2013). However, MeHg levels in paddy soil are not only determined by the methylation of IHg, but also by the rate of demethylation of MeHg. A recent study showed that non-Hg

methylating bacteria such as *Xanthomonadales* and *Chitinophagaceae* were strongly correlated with soil MeHg contents (Liu et al., 2019b).

2) To determined how IHg and MeHg are absorbed and translocated into the rice plants. The pathways and regulating genes for uptake and translocation of Cd and As have been well demonstrated (Atwell et al., 2014; Chen et al., 2019; Hu et al., 2016; Huang et al., 2018; Liu et al., 2004a; Luo et al., 2018; Mei et al., 2009; Zhang and Wing, 2013). However, little is known about the uptake, translocation and regulating genes of IHg and MeHg in rice plants (Meng et al., 2012, 2014a).
3) To understand how farming activities affect Hg migration and transformation in paddy fields. As discussed in this review, application of fertilizer, crop rotation, water management, and other management practices can lead to changes in soil properties and nutrient uptake by plants, both of which can affect Hg levels and transformation in rice plants. Additionally, what are the effects of cultivation density and growth conditions of rice plants on photochemical processes in Hg transformation in paddy soil.
4) To assess whether there is potential MeHg exposure risks in paddy fields with low background levels of Hg. Most studies have been focused on MeHg in rice plants grown in soils with high Hg levels. The global mobility of Hg could result in MeHg exposure to the relatively low Hg level paddy fields. Several recent studies suggested that returning rice straw could significantly increase the MeHg concentration in paddy soil and rice grain. Concentrations of MeHg in grain from paddies practicing the return of rice straw were 8.4-fold greater than grain from paddies not untreated with rice straw (Liu et al., 2016b; Tang et al., 2018, 2019).

The above are examples of future research needs. The focus of this review has been on the transformation of Hg in paddy soil, uptake and translocation of Hg in rice plants, and effects of farming activities. Since plants can absorb IHg and MeHg from surrounding air, these Hg translocation processes may also contribute significantly to potential accumulation of Hg in rice plants. Clearly, targeted research effort are needed to thoroughly understand Hg transformations in the environment so that management strategies may be developed to systematically reduce Hg exposure risks in the rice paddy system.

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Declaration of competing interest

The authors declare no competing interests.

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